

React & Modern JavaScript

Build powerful single-page applications with React, hooks, and the modern JS ecosystem.

Welcome to the course! This document contains all the materials, lessons, and exercises you will need to successfully complete the curriculum. We recommend reading through this guide carefully and taking notes as you progress.

Category: Development | Level: Intermediate | Est. Time: 12 Hours

Course Objectives & Overview

By the end of this course, you will have a solid understanding of the core concepts, practical applications, and best practices associated with this topic. This structured material is designed to guide you step-by-step from foundational theories to advanced problem-solving techniques.

Ensure you complete all practical exercises and review the lesson summaries before attempting the final exam.

Lesson 1: React Fundamentals

React is a JavaScript library for building user interfaces.

- JSX syntax and Babel transpilation
- Creating and nesting components
- Props: passing data down to child components
- State with useState hook
- Rendering lists with .map() and key prop
- Handling events in React
- Conditional rendering
- Component lifecycle and useEffect
- Basic form handling with controlled inputs

Key concept: React uses a virtual DOM to efficiently update only the parts of the UI that change.

Key Takeaways:

- Review the main concepts outlined above.
- Practice the techniques discussed.
- Write down any questions for further research.

Lesson 2: Hooks & State Management

React hooks make it easy to add state and side effects to function components.

- `useState` for local component state
- `useEffect` for side effects (data fetching, subscriptions)
- `useContext` for sharing state without prop drilling
- `useReducer` for complex state logic
- Custom hooks: extracting reusable logic
- React Context API for global state
- Introduction to Zustand / Redux Toolkit

Key concept: Hooks let you use state and React features in functional components without writing a class.

Key Takeaways:

- Review the main concepts outlined above.
- Practice the techniques discussed.
- Write down any questions for further research.

Conclusion & Next Steps

Congratulations on reaching the end of the written material for this course! Your next step is to log into the platform and take the final assessment. Review your personal notes, ensure you understand the key takeaways from each lesson, and apply what you have learned.

Good luck!